Nearby Evidence Shapes How Transformers Respond to Planted Errors in Their Own Reasoning

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Abstract

Can an autoregressive language model recover from a planted error in its own reasoning trace? Long-horizon reasoning is fragile because a false intermediate step can become part of the context the model must reason from. Simple compounding-error arguments turn this concern into a strong assumption: once a step is false, later reasoning cannot recover. We test a different possibility. Recovery may depend less on falsehood itself than on local falsifiability: whether the current proof state contains cheap counterevidence. We instantiate this test by intercepting proofs the model already solves, planting a single audited false step at controlled positions and proof-search costs, and formally validating the continuation. Across models from 1.5B to 32B parameters, nearby contradictions often elicit doubt and repair, whereas errors requiring broader proof search are often silently absorbed and reused. The result is a falsifiability gradient: per-step errors do compound, but primarily when the current context provides no cheap counterevidence. This refines the compounding-error debate and gives a practical diagnostic for reasoning traces: the riskiest errors are not merely wrong, but wrong where the surrounding context gives the model little reason to doubt.

1 Introduction

Autoregressive language models generate one token at a time, so long-horizon reasoning can be fragile. Recent theory frames this fragility as an intrinsic stability limit of single-path autoregressive execution, not merely a byproduct of combinatorial search: small epistemic perturbations can erode decision advantage with execution length, and CoT perturbation studies show heterogeneous vulnerabilities across error types (Liao, 2026; Aravindan and Kejriwal, 2026). A standard compounding argument says that if each step is correct with probability $(1 - \varepsilon)$ and a wrong step can never be repaired, then the chance that a T-step derivation is fully correct decays like $(1 - \varepsilon)^T$ (Dziri et al., 2023). This geometric sketch is only one topology of accumulation: off-policy sequential prediction can incur $O(\varepsilon T^2)$ mistakes under induced-state mismatch (Ross et al., 2011), and recent CoT audits identify late-stage fragility, where late errors can be more damaging because the trace provides less logical runway for redundant routing or self-correction (Zhang et al., 2025; Neumann and Singh, 2025). The compounding argument therefore relies not just on independence and a constant error rate, but on unrecoverability; that last premise carries much of the force.

The intrinsic question is not just whether a trace contains an error, but when the model treats that error as something to doubt. Recent self-correction work shows that apparent doubt can depend on chat-template role labels, while probabilistic soundness frameworks evaluate each step against previously verified premises rather than using holistic judges (Chen et al., 2026a; You et al., 2025). Downstream systems also consume and route model-generated traces: trace disagreement can serve as a knowledge-representation signal, diverse paths can be routed during distillation, and visual diagnosis tools expose premise-level error propagation (Wawer and Chudziak, 2026; Lei et al., 2025; Chen et al., 2026b). We therefore study the unassisted rollout: if we plant one error in a model's own proof, does the model build on the error, route around it, or complete a valid proof anyway?

Prior work gives reasons to expect both fragility and robustness. Interventional studies perturb visible chain-of-thought and report surprising robustness (von Recum et al., 2026); mechanistic analyses find redundant, self-repairing structure inside networks (McGrath et al., 2023; Lad et al., 2024). A parallel architectural response makes robustness structural: recursive models use unbounded recursion to isolate active sub-contexts while separating local attention from global memory (Yang et al., 2026), and horizon-reduction work treats effective horizon length as a stability bottleneck whose reduction enables horizon generalization (Kim et al., 2026). These systems form the structural-governance backdrop for our question: they change the execution structure, whereas we ask what happens inside a single visible rollout. None of these lines fully settles our question, because two details matter here: whether the planted step is actually false, and whether apparent recovery is a genuine proof or a restatement of a prompted goal. Once those are controlled, the central pattern is simple: the model avoids errors when nearby evidence refutes them, and uses planted facts more often when detection requires broader proof search. In the expanded Qwen2.5-7B sweep, globally checkable falsehoods yield closure-valid completion rates of 0.360–0.447 across injection positions and injection-dependence rates of 0.233–0.420, while one-hop falsehoods remain 0.633–0.740 closure-valid and produce much higher verbalized doubt.

Contributions.

1. **Perturb-and-trace protocol.** We plant a controlled error in a model's own proof and use a formal validator to classify closure-valid completion, injection-dependence, goal restatement, or derailment. This distinction matters because final-answer accuracy can count parroted goals as recovery.
2. **Truth-status audit.** We audit every planted step, preventing off-path-but-entailed facts from being counted as errors.
3. **Nearby-evidence conditions.** We compare planted steps that range from meaning-preserving paraphrases to errors contradicted by nearby statements and errors detectable only from the full rule closure. All rates and confidence intervals are generated from archived result artifacts.
4. **Controls and limitations.** Certificate and polarity-control experiments support a role for local evidence but also expose boundary conditions: local certificates increase explicit doubt and reduce injection-dependence, while the strict lexical-polarity design fails to produce a clean local false positive categorical cell in this grammar.

2 Setup

Evaluation domain. We use PrOntoQA-OOD (Saparov and He, 2023), which generates fictional ontologies (e.g., "every wumpus is a tumpus") and asks the model to derive a target proof. Its rigid grammar lets a proof validator parse each sentence. We use a fixed PrOntoQA-OOD cohort with multi-hop proofs.

Model cohort. We evaluate six open-weights models to isolate scale, architectural lineage, and reasoning-specific training: Qwen2.5-Instruct (1.5B, 7B, 32B), OLMo-2-7B-Instruct, Llama-3.1-8B-Instruct, and DeepSeek-R1-Distill-Qwen-7B. All models are run greedily and without fine-tuning, with Qwen2.5-7B serving as our primary subject. We perturb only proofs that the model successfully solves and that pass the grammar validator. Because every injected trajectory starts from a successful proof, the reported closure-valid completion rates are upper bounds.

The perturb-and-trace protocol. The model first writes a correct proof. We then intercept the proof, select one intermediate step (early, middle, or late), and overwrite it with a planted sentence. The model resumes generation with the same token limits, no external feedback, and no prompt asking it to self-correct. We record the full continuation and validate what it writes next. For latent comparisons, we also sample eight temperature-0.8

continuations from the same interception point with the correct step left intact, retaining only valid continuations. Hidden states are cached at five depths throughout the sequence.

Planted-step conditions. The independent variable is the local verification cost of the planted step. We use "falsifiability gradient" to mean the distance, in local proof context and rule closure, between a planted claim and the nearest evidence that refutes it. We structure six perturbations into this gradient, ordered by how easily the step can be checked against the surrounding proof:

(a) **Benign paraphrase**: The correct step, reworded. Controls for the baseline cost of interrupting the generation trace.

(b) **Off-path category (True interruption)**: A fact absent from the gold proof path but logically entailed by the world premises.

(c) **Distractor**: A fabricated rule entirely irrelevant to the active entity.

(d) **Contradiction**: The negation of a fact already stated in the proof. This is false and conflicts with adjacent context at distance zero.

(e) **Negated step**: The negation of the step about to be derived. This is false and conflicts with no prior statement, but is refutable via a single-hop inference from the surrounding proof.

(f) **Globally-checkable falsehood**: A categorical claim that is strictly unentailed. It lacks nearby conflicting evidence and is detectable only by searching the entire rule closure.

2.1 Auditing Perturbation Validity

In entailment-dense PrOntoQA worlds, a fact absent from the gold proof can still be true. We therefore audit every planted atom against the rule closure and report only families with verified truth status. Because globally-checkable falsehoods are rarer under the constrained generator, the primary sweep expands that family to 150 examples per injection position and matches it with true-interruption, benign-paraphrase, and one-hop-falsehood controls (Table 1).

Condition	n	Closure-valid [95% CI]	Inj.-dep. [95% CI]	Doubt [95% CI]	Unparsed
true interruption	450	0.856 [0.818,0.889]	0.000 [0.000,0.000]	0.020 [0.007,0.036]	0.013
benign paraphrase	450	0.849 [0.813,0.882]	0.000 [0.000,0.000]	0.080 [0.056,0.107]	0.024
1-hop falsehood	450	0.671 [0.624,0.722]	0.016 [0.004,0.029]	0.336 [0.291,0.387]	0.038
global falsehood	450	0.393 [0.342,0.442]	0.342 [0.293,0.384]	0.013 [0.004,0.024]	0.018

Table 1: Expanded audited primary PrOntoQA perturbation set. Each condition pools three injection positions (150 examples per position); confidence intervals are problem-cluster bootstrap intervals.

2.2 Metrics for Completion and Failure

Because a model can print the correct final line via heuristic shortcuts while producing an invalid proof, final-answer accuracy can overstate recovery. We therefore formally parse and validate the proof trace itself.

A continuation is classified as a **closure-valid completion** if every stated fact follows from the true premises and prior valid steps under the rule closure, culminating in the target. This metric measures semantic validity under closure, not faithful replay of the original derivation. We categorize failure modes as follows:

Injection-dependent: Subsequent reasoning follows only if the planted atom is assumed as true. For genuinely false, positive-category injections, this constitutes strict poisoning.

Parroted: The target is reached, but the proof contains an unentailed intermediate step.

Derailed: The continuation ends on an incorrect statement.

To quantify explicit error-flagging, we scan for hedging and self-correction tokens (e.g., "wait," "however," "contradict"). This heuristic detector serves strictly as a behavioral flag for doubt and is not treated as proof of logical repair.

Validator Limitations: The proof validator is closure-based: it checks whether each fact follows from the true premises and prior valid steps, but it allows the model to skip steps. We found that closure-valid continuations can include derivation-free jumps to the goal; at late positions, the proof has fewer intermediate steps left to skip. Furthermore, strict poisoning is only mechanically detectable for positive category claims, as the grammar derives nothing from a negation. Thus, low strict-poisoning rates reported for negation families represent a mechanical measurement boundary rather than definitive evidence of model resilience. Finally, because discourse markers are stripped before parsing, we exclude and separately count unparsed runs to account for instances where the model breaks the parser by copying injected phrasing.

Example. Table 2 shows one audited PrOntoQA instance. The relevant chain is *Stella is a lorpus* $\rightarrow$ *Stella is a shumpus* $\rightarrow$ *Stella is a wumpus* $\rightarrow$ *Stella is not liquid*; the full prompt also contains distractor facts and a second premise (*Stella is a dumpus*). The same world produces all four validator outcomes depending on the planted step and the continuation.

Condition	Planted step	Continuation excerpt	Validator
1-hop falsehood	Stella is not mean.	"Contradiction, but not needed for proof. Stella is a dumpus. Every dumpus is not opaque. . . . Stella is not liquid."	closure-valid
global falsehood	Stella is an impus.	"Every impus is not red. Stella is not red. . . . Stella is a wumpus. Every wumpus is a sterpus. . . . "	injection-dependent
benign paraphrase	Thus, Stella is not opaque.	The model eventually states the target, but inserts underivable category steps on the way.	parroted
1-hop falsehood	Stella is not a dumpus.	The model continues with other facts and never ends on *Stella is not liquid*.	derailed

Table 2: A single instance illustrating the validator categories. For genuinely false, positive-category injections, injection-dependence is strict poisoning.

3 Results

3.1 Nearby Evidence and Recovery

Table 3 presents the primary outcomes regenerated from validated artifacts. Because any interruption can lower closure-valid completion, we interpret all perturbations against the benign-paraphrase baseline. The central contrast is whether the surrounding proof contains nearby evidence against the planted error. Errors contradicted by nearby statements are more often flagged or avoided; errors requiring a broader proof search are more often used in later steps. The table also shows exceptions and parser failures, so the result is evidence for this nearby-evidence pattern rather than proof of a single mechanism.

Errors requiring broader proof search. When a planted falsehood is detectable only by checking the full rule closure, the model often completes a confident-looking proof and uses the planted fact in later derivation steps, with little explicit doubt.

Errors with nearby evidence against them. When a falsehood can be checked against adjacent context, explicit doubt is largest among the main families, while closure-valid completion remains substantially above the global falsehood family. The model's response is associated with how easy the error is to check against the surrounding proof, not merely with whether it is false.

Problem-clustered robustness. The same pattern survives a fixed-effect logistic model fit to the raw Qwen2.5-7B validation rows with injection position included and problem IDs resampled as clusters. Relative to benign paraphrase, both one-hop falsehoods and errors requiring broader proof search reduce the adjusted odds of closure-valid completion,

Family	Inj.	n	Closure-valid [95% CI]	Inj.-dep.	Parroted	Doubt	Unparsed
benign paraphrase	early	150	0.747 [0.672,0.810]	—	0.027	0.087	0.053
benign paraphrase	mid	150	0.880 [0.818,0.923]	—	0.040	0.127	0.020
benign paraphrase	late	150	0.920 [0.865,0.954]	—	0.000	0.027	0.000
true interruption	early	150	0.820 [0.751,0.873]	0.000	0.127	0.007	0.020
true interruption	mid	150	0.867 [0.803,0.912]	0.000	0.100	0.027	0.013
true interruption	late	150	0.880 [0.818,0.923]	0.000	0.100	0.027	0.007
distractor rule	early	130	0.731 [0.649,0.800]	0.000	0.100	0.023	0.054
distractor rule	mid	130	0.715 [0.633,0.786]	0.000	0.138	0.031	0.092
distractor rule	late	130	0.762 [0.681,0.827]	0.000	0.146	0.031	0.054
contradiction (dist. 0)	early	130	0.731 [0.649,0.800]	0.000	0.108	0.231	0.077
contradiction (dist. 0)	mid	130	0.792 [0.715,0.853]	0.000	0.023	0.269	0.100
contradiction (dist. 0)	late	130	0.854 [0.783,0.904]	0.008	0.023	0.177	0.069
false, 1-hop checkable	early	150	0.633 [0.554,0.706]	0.000†	0.147	0.467	0.047
false, 1-hop checkable	mid	150	0.640 [0.561,0.712]	0.020†	0.213	0.273	0.013
false, 1-hop checkable	late	150	0.740 [0.664,0.804]	0.027†	0.087	0.267	0.053
false, globally checkable only	early	150	0.447 [0.369,0.527]	0.233	0.233	0.007	0.027
false, globally checkable only	mid	150	0.360 [0.288,0.439]	0.373	0.193	0.013	0.013
false, globally checkable only	late	150	0.373 [0.300,0.453]	0.420	0.173	0.020	0.013

Table 3: Formally validated outcomes for the main Qwen2.5-7B PrOntoQA sweep. Cell sizes, rates, and confidence intervals are regenerated from validated result artifacts; rows with $n = 150$ use the expanded audited sample, while the distractor and contradiction rows retain the original auxiliary sweep. "Closure-valid" is semantic validity under rule closure, not strict step-by-step replay. "Inj.-dep." (injection-dependent) indicates that a subsequent reasoning step logically follows only if the planted atom is assumed to be true; for genuinely false positive-category claims, this constitutes strict error poisoning. †For negated injections, this validator flag is not strict poisoning.

with the global falsehood family showing the larger reduction. For explicit doubt, the direction reverses when nearby evidence is available: contradiction and one-hop falsehood increase doubt, while global falsehoods remain below the benign baseline. Intervals are problem-cluster bootstrap intervals stored in the cluster-statistics artifact.

A Dose-Response in Inferential Distance. The availability of nearby evidence can be varied parametrically. By negating a fact the model will derive k hops downstream, we place the contradicting evidence exactly k inferences away. At early injection positions, explicit doubt and closure-valid completion decline as the evidence moves farther away, then plateau for longer chains. At the 32B scale, the curve shifts upward and compresses, indicating that scale lifts absolute robustness without removing the distance dependence.

Verification prompting does not reliably fix errors without nearby evidence. Explicitly instructing the model to verify facts against the rules before using them fails to mitigate the dangerous case in the available verification run. At the middle injection position, globally-checkable falsehoods still show injection-dependence 0.371 and closure-valid completion 0.343 ($n = 35$), with only 0.029 verbalized doubt. By contrast, one-hop falsehoods under the same prompting condition produce 0.423 doubt ($n = 130$). The instruction raises visible caution most clearly when the lie can already be checked against nearby context; it is easier to induce cautionary language than to induce checking of claims with no nearby evidence against them.

3.2 Targeted Artifact-Locked Controls

Following the primary sweep, we executed two strictly paired, artifact-locked control experiments to probe the role of local evidence and rule out potential lexical confounds.

Local-Certificate Flip. We tested whether artificially injecting local evidence changes the model's response to an otherwise global falsehood. This intervention inserts a local "certificate" that directly contradicts the planted global claim. Across 100 paired triples at each injection position (183 problem clusters), each condition contributes 300 continuations. The presence of the certificate lowered injection-dependence ($\Delta = -0.127$) and strongly increased explicit doubt ($\Delta = +0.370$). However, it did not guarantee full proof repair: closure-valid completion improved only marginally ($\Delta = +0.090$), accompanied by an increase in parser-breaking formatting. The certificate primarily changes error-flagging behavior and reduces injection-dependence; it does not cleanly improve closure-valid completion.

Condition	n	Closure-valid [95% CI]	Inj.-dep. [95% CI]	Doubt [95% CI]	Unparsed
global baseline	300	0.377 [0.317,0.436]	0.153 [0.112,0.201]	0.010 [0.000,0.023]	0.040
local certificate	300	0.467 [0.402,0.538]	0.027 [0.010,0.043]	0.380 [0.319,0.445]	0.317
irrelevant certificate	300	0.500 [0.436,0.564]	0.240 [0.187,0.298]	0.023 [0.010,0.041]	0.037

Table 4: Local-certificate flip. The local certificate makes falsifying evidence immediately available before the same globally-checkable falsehood; the irrelevant certificate is length-matched and true but does not locally falsify the falsehood. Confidence intervals are problem-cluster bootstrap intervals.

Lexical-Polarity Control. To isolate logical locality from mere lexical polarity (affirmative versus negative phrasing), we attempted a 2×2 locality-by-polarity sweep. The rigid ontology grammar prevented a clean local, affirmative false positive without changing the category rules, bottlenecking the fully matched cohort to 7 problems (84 validated rows). While underpowered, the available data align explicit doubt with local evidence rather than polarity. The closure-valid signal, however, remains too noisy for formal inference.

Condition	n	Closure-valid [95% CI]	Doubt [95% CI]	Inj.-dep. [95% CI]	Unparsed
local positive	21	0.714 [0.476,0.905]	0.286 [0.095,0.476]	—	0.095
local negative	21	0.524 [0.286,0.762]	0.333 [0.095,0.571]	—	0.048
global positive	21	0.667 [0.429,0.905]	0.000 [0.000,0.000]	0.238 [0.048,0.476]	0.000
global negative	21	0.714 [0.524,0.905]	0.143 [0.000,0.333]	—	0.000

Table 5: Lexical-polarity control. The strict categorical local-positive cell was infeasible in this grammar; the generated local-positive cell uses affirmative attributes, so the table is reported as a boundary-condition check rather than a decisive lexical control. Confidence intervals are problem-cluster bootstrap intervals.

3.3 Strict Stepwise Validation

The primary proof validator is closure-based: it checks whether every generated fact follows from true premises and prior valid steps, but allows skipped intermediate steps. The lexical-polarity control additionally stores a strict replay validator that asks whether the continuation follows the original gold suffix step by step. This stricter audit is reported separately in Table 6; crucially, it never overwrites the closure validator used for our headline metrics.

Control	n	Strict replay?	Closure-valid	Strict final target match	Strict final mismatch
Lexical-polarity control	84	yes	55 (0.655)	66 (0.786)	18 (0.214)
Local-certificate flip	900	no	403 (0.448)	—	—

Table 6: Strict stepwise validation audit for new controls. Counts are shown with rates in parentheses. The strict replay audit records whether the continuation reaches the stored final target under the suffix comparison; it is not substituted for the closure validator used in headline metrics and is reported only where implemented.

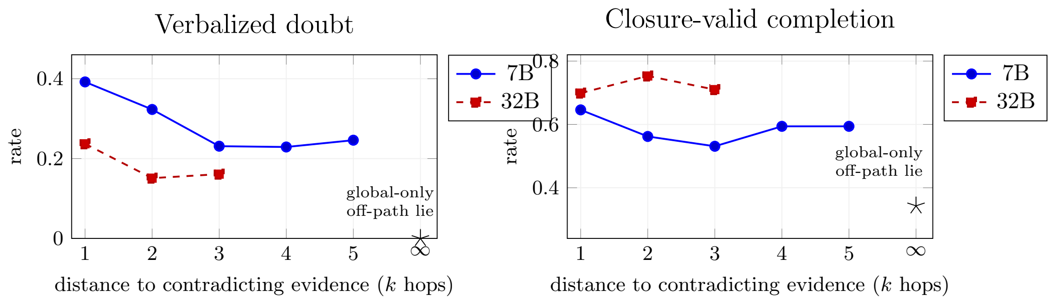

Figure 1: **The response to a planted lie is a dose–response in inferential distance.** Negating the fact the model will derive k hops downstream places the falsifying evidence k inferences from local context. Both verbalized doubt (left) and closure-valid completion (right) decline as k grows and then plateau ($k \geq 3$); globally-checkable-only lies (off-path, $k \to \infty$, $\star$) are often used without explicit flagging. 32B shifts the curve upward rather than removing the distance pattern.

3.4 Task Derivation Structure Dictates Recoverability

We applied the same mid-trajectory corruption protocol across three distinct task structures. The pattern tracks the logical redundancy of the task. In PrOntoQA, multiple derivation paths can reach the goal, so a planted error can sometimes be routed around. In chained arithmetic, every intermediate value is load-bearing and there are no redundant paths, so an error that the model uses almost always propagates to the next step.

This comparison clarifies what "easy to check" means here. In GSM8K, the corrupted value is technically checkable by recomputation, yet Table 7 shows that it is usually used. The PrOntoQA result depends on facts explicitly visible in the context window, not on arithmetic the model must spontaneously redo. Models appear more likely to re-check information they can re-read than information they must recompute, so recoverability depends on task structure as much as model architecture.

Task	Closure-valid	Inj.-dep.	n
PrOntoQA (redundant derivations)	0.360	0.373*	150
GSM8K (natural-language math)	0.088	0.877	57
Chained arithmetic (no redundancy)	0.020	0.990	99

Table 7: Mid-trajectory corruption across task structures. Numbers are regenerated from the task summary artifacts; the PrOntoQA row uses the expanded globally-checkable falsehood cell. *In PrOntoQA, absorption is measured as injection-dependence under the proof validator; in arithmetic tasks, absorption corresponds to propagation of the corrupted value.

3.5 Goal-Anchor Copying Does Not Fully Explain Recovery

To rule out the possibility that apparent recovery is merely the model re-reading and copying the target statement provided in the prompt, we conducted two targeted ablations. First, we obscure the target's truth value by reframing each problem as a boolean query ("True or false: s?", where s is the true target in half the instances and its negation in the other). This exposes the statement but removes the definitive answer to copy. Under this constraint, closure-valid completion is attenuated but persists in the available pilot data, showing that goal-copying heuristics cannot fully explain the error-recovery phenomenon.

Second, we applied a direct attention mask to the goal span. This intervention collapsed closure-valid completion across both the falsehood and benign arms. Because masking the goal removes the model's structural stopping condition, this manipulation fundamentally breaks the task rather than isolating the recovery mechanism, yielding a negative result.

However, it provides one critical qualitative insight: goal access facilitates overall derivation quality, rather than specifically driving error repair.

3.6 Latent Signatures and the Stylistic Confound

We investigated whether error recovery leaves a detectable signature in hidden states and found a strongly negative result. Jensen–Shannon divergence (ρ_h^{func}) between perturbed and gold next-token distributions spikes immediately after the injection, but fails to track semantic content. The benign paraphrase — a strictly stylistic edit — produced the largest early divergence of any perturbation family. Because the model adopts the injected phrasing for the rest of its proof, the metric registers surface style as heavily as a planted logical lie.

Subsequent controls confirmed this is a geometric confound rather than a sampling artifact. Final-layer cosine divergence remained statistically indistinguishable between benign paraphrases and verified falsehoods. As instrumented, trace-based divergence cannot distinguish a logical error from a stylistic change. Our primary claims therefore remain strictly behavioral and formally validated.

3.7 Scale and Lineage: Explicit Doubt and Recovery Dissociate

Scaling Qwen2.5 from 1.5B to 32B suggests that scale improves recovery from errors with no nearby evidence against them and reduces how often the continuation uses the planted error, but it does not make explicit doubt a reliable proxy for closure-valid completion.

	globally-checkable falsehood			1-hop falsehood
Model	closure-valid	inj.-dep.	doubt	doubt
Qwen2.5-1.5B	0.154–0.462	0.308–0.615	0.000–0.000	0.000–0.018
Qwen2.5-7B (primary)	0.360–0.447	0.233–0.420	0.007–0.020	0.267–0.467
Qwen2.5-32B	0.600–0.733	0.100–0.367	0.033–0.067	0.129–0.280

Table 8: Scale sweep over Qwen2.5 models. Ranges are over early, middle, and late injection positions and are regenerated from summary artifacts. The Qwen2.5-7B row uses the expanded primary artifact, making it numerically comparable to Table 3; the other size rows use the matched cross-model summary artifacts.

The same nearby-evidence pattern appears across architectural lineages. Both OLMo-2-7B-Instruct and Llama-3.1-8B-Instruct show the behavioral gradient in the cross-model summary artifacts. Crucially, both models can recover from errors with nearby evidence against them while showing little explicit doubt, indicating that successful error repair does not require the model to explicitly flag the mistake.

Finally, evaluating DeepSeek-R1-Distill-Qwen-7B isolates the impact of reasoning-specific reinforcement learning. While closure-valid completion remains broadly in the same range as its base model, explicit flagging of one-hop falsehoods is much higher. Across the model set, explicit flagging changes more than closure-valid completion. We therefore treat explicit doubt as a trainable interface behavior and an unreliable proxy for closure-valid completion.

4 Limitations

Our coverage is still narrow: two synthetic families plus GSM8K, deterministic decoding, one null-sampling seed, and a model set that remains small and Qwen-heavy despite including OLMo-2, Llama-3.1, and R1-Distill. A formal-verification task such as Lean is the natural next regime. Several design limitations also remain. The globally-checkable falsehood family is generation-constrained, although the primary sweep expands the matched validated sample to 150 examples per injection position. The lexical-polarity control does not cleanly isolate polarity; the prompted target is only partly addressed by parroting and goal-hidden QA controls; our latent/geometric metrics are negative results; and closure-based validation

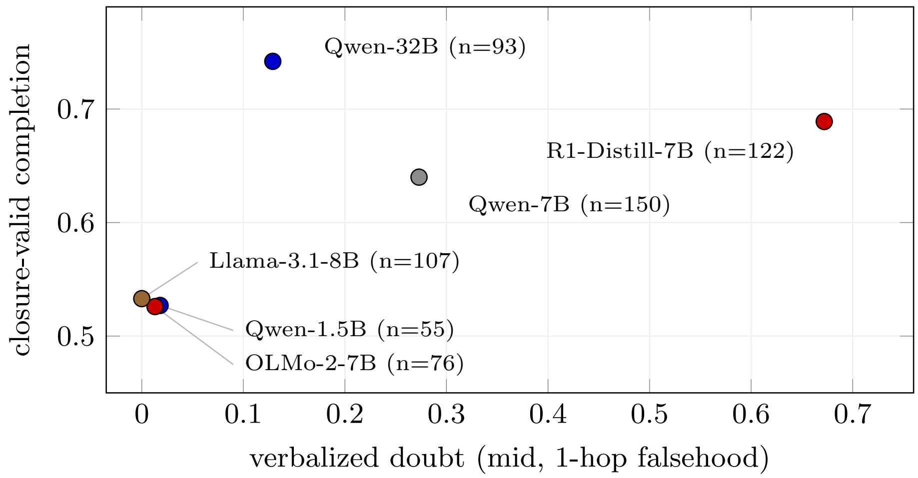

Figure 2: **Explicit doubt and recovery dissociate.** Across six models (1-hop falsehood, mid injection), verbalized doubt varies much more than closure-valid completion. Points are cell rates with the per-model n printed on the labels. The Qwen-7B point uses the expanded primary artifact; the other points use the matched cross-model summaries. Reasoning-RL training (R1-Distill) increases explicit flagging at flat closure-valid completion; the base/instruct models recover at comparable rates while saying little. Verbalizing a caught error and correcting it are dissociable behaviors.

permits step-skipping, so the strict replay audit should not be conflated with closure-valid completion.

5 Related work

CoT interventions and error recovery. Perturbation studies of visible CoT report behavioral robustness, early-intervention effects, and doubt/length changes (von Recum et al., 2026); faithful/unfaithful recovery annotations precede our closure-valid/parroting split (Yee et al., 2024). Our delta is automated formal validation, semantic error control, a per-injection truth-status audit, and the nearby-evidence comparison.

Faithfulness, self-correction, and compounding. Lanham et al. (2023); Turpin et al. (2023); Zhang et al. (2023); Huang et al. (2024) show that stated reasoning can be unfaithful, self-correction can be weak, and models may commit to mistakes they can separately recognize. Our single-pass setting locates when that recognition is deployed in-rollout. Mechanistic self-repair work (McGrath et al., 2023; Lad et al., 2024), trajectory geometry (Sun et al., 2026), and compounding analyses (Dziri et al., 2023) provide the surrounding frame: robustness machinery can survive some perturbations, but the price depends on whether task structure supplies nearby counterevidence (§3.4).

6 Conclusion

Single-pass autoregressive reasoning is neither uniformly brittle nor reliably self-correcting. In our controlled setting, recovery depends partly on whether the planted error can be checked against nearby context. Nearby contradictions are often survived; broader-search errors are often reused silently. Thus, compounding arguments fit best when the proof lacks cheap counterevidence and overstate fragility when contradiction is local. Evaluations should track local refutability, not just falsity. For systems that rely on model-generated traces, this makes counterevidence a first-class part of the reasoning state. The risky errors are not merely wrong; they are wrong in ways the current context does not make cheap to doubt.

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